



9601 - The Star Formation Histories of Isolated Low-Mass Galaxies: Setting a New Empirical Benchmark

Cycle: 5, Proposal Category: GO

INVESTIGATORS

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OBSERVATIONS

<i>Folder</i>	<i>Observation</i>	<i>Label</i>	<i>Observing Template</i>	<i>Science Target</i>
Sextans B				
	1	SexB	NIRCam Imaging	(1) SEX-B
Antlia				
	2	Antlia	NIRCam Imaging	(2) ANTLIA-DWARF-GALAXY
Antlia B				
	4	AntliaB	NIRCam Imaging	(4) Antlia-B
UGC4879				
	3	UGC4879	NIRCam Imaging	(3) UGC-4879

ABSTRACT

The most detailed and precise measurements on the stellar mass assembly and evolution of low-mass galaxies come from systems in close proximity to the Milky Way and M31, as these have been the galaxies with the best observations. Yet, it is well-established that low-mass galaxies are easily transformed by environment and that the evolution of satellite dwarfs has been heavily influenced by proximity to a massive galaxy. As this is a highly biased sample, there are limitations in using these low-mass satellites as our primary sample to test physical models of the galaxy assembly process. Even to simply understand the full history of the population of LG dwarfs, the effects of environment on the formation and evolution of low-mass nearby galaxies must first be disentangled.

We propose to derive the ancient star formation histories of 4 isolated low-mass galaxies that are far from a massive host (>1 Mpc) using deep JWST imaging of the resolved stellar populations. These observations will complete the sample of isolated dwarfs with robust SFHs within 1.6 Mpc. JWST makes SFHs for these galaxies possible for the first time (and recent work on Leo P has proved that). We will measure the stellar mass assembly of these island universes across all cosmic timescales and test whether the newly emerging pattern of paused and reignited star formation noted among the few isolated dwarfs with the requisite data is common. Without the complicating factor of a nearby massive galaxy woven through their history, these isolated galaxies offer an unparalleled view of galaxy formation and secular evolution from cosmic dawn to the present-day.

OBSERVING DESCRIPTION

We will image each galaxy with NIRCcam with the short-wavelength filters F090W and F150W, with simultaneous imaging in the long-wavelength filters F330M and F335M. Our science goals require that the SW imaging reach below the old main sequence turn-off with a $\text{SNR} \sim 10$ for a metal-poor, 13 Gyr old star. We have used a MEDIUMDEEP8 readout and a subpixel dither pattern to mitigate cosmic rays and bad pixels, improve PSF sampling, and reduce flat-field errors thereby maximizing the SNR. We do not intend to cover the chip gaps as uniform depth is preferred for the imaged stellar fields. We have selected 6-8 groups per integration. We selected orient constraints to ensure the NIRCcam field of view optimally covers the stellar disks of the galaxies while avoiding any bright nearby stars. We request NIRISS imaging in parallel to assess possible foreground contamination and with the potential of detecting an extended stellar halos in the galaxies.

Proposal 9601 - Targets - The Star Formation Histories of Isolated Low-Mass Galaxies: Setting a New Empirical Benchmark

Fixed Targets	#	Name	Target Coordinates	Targ. Coord. Corrections	Miscellaneous
	(1)	SEX-B	RA: 10 00 3.1950 (150.0133125d) Dec: +05 21 10.20 (5.35283d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Galaxy</i> <i>Description=[Dwarf irregular galaxies, Field galaxies]</i>		
	(2)	ANTLIA-DWARF-GALAXY	RA: 10 04 4.8973 (151.0204054d) Dec: -27 21 11.21 (-27.35311d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Galaxy</i> <i>Description=[Dwarf irregular galaxies, Field galaxies]</i>		
	(3)	UGC-4879	RA: 09 15 55.5387 (138.9814113d) Dec: +52 49 38.05 (52.82724d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Galaxy</i> <i>Description=[Dwarf galaxies, Field galaxies]</i>	Epoch of Position: 2015.5	
	(4)	Antlia-B	RA: 09 48 57.0060 (147.2375250d) Dec: -25 57 7.57 (-25.95210d) Equinox: J2000 <i>Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.</i> <i>Category=Galaxy</i> <i>Description=[Dwarf galaxies, Field galaxies]</i>	Epoch of Position: 2000	

Proposal 9601 - Observation 1 - The Star Formation Histories of Isolated Low-Mass Galaxies: Setting a New Empirical Benchmark

Observation	Proposal 9601, Observation 1: SexB										Tue Apr 14 18:00:09 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCам Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnostics	(SexB (Obs 1)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.											
	(Visit 1:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous		
	(1)	SEX-B	RA: 10 00 3.1950 (150.0133125d) Dec: +05 21 10.20 (5.35283d) Equinox: J2000									
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	Category=Galaxy Description=[Dwarf irregular galaxies, Field galaxies]											
Template	NIRCам Imaging					NIRISS Imaging						
	Module: ALL											
	Subarray: FULL											
	Target Placement: Module gap (large extended source)											
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	NONE						1		4-POINT-SMALL-WITH-NIRISS		NO_DITHERING
Spectral Elements	NIRCам Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F300M	MEDIUMDEEP8	7	3		12	4	12712.336		
	2	F150W	F335M	MEDIUMDEEP8	8	3		12	4	14644.954		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F090W		NIS	9	8		4	32	12712.336		
	2	F150W		NIS	9	8		4	32	12712.336		

Proposal 9601 - Observation 1 - The Star Formation Histories of Isolated Low-Mass Galaxies: Setting a New Empirical Benchmark

Special Requirements	Aperture PA Range 109.92542306 to 124.42542306 Degrees (V3 110.0 to 124.5) No Parallel Attachments
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Proposal 9601 - Observation 2 - The Star Formation Histories of Isolated Low-Mass Galaxies: Setting a New Empirical Benchmark

Observation	Proposal 9601, Observation 2: Antlia										Tue Apr 14 18:00:09 GMT 2026	
	Diagnostic Status: Warning											
	Observing Template: NIRCам Imaging											
	Coordinated Parallel Template(s): NIRISS Imaging											
Diagnostics	(Antlia (Obs 2)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.											
	(Visit 2:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.											
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections		Miscellaneous			
	(2)	ANTLIA-DWARF-GALAXY	RA: 10 04 4.8973 (151.0204054d) Dec: -27 21 11.21 (-27.35311d) Equinox: J2000									
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.											
	Category=Galaxy Description=[Dwarf irregular galaxies, Field galaxies]											
Template	NIRCам Imaging					NIRISS Imaging						
	Module: ALL											
	Subarray: FULL											
	Target Placement: Module gap (large extended source)											
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes
	1	NONE						1		4-POINT-SMALL-WITH-NIRISS		NO_DITHERING
Spectral Elements	NIRCам Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID	
	1	F090W	F300M	MEDIUMDEEP8	6	3		12	4	10779.717		
	2	F150W	F335M	MEDIUMDEEP8	7	3		12	4	12712.336		
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID	
	1	F090W		NIS	8	7		4	28	9920.775		
	2	F150W		NIS	8	7		4	28	9920.775		

Proposal 9601 - Observation 2 - The Star Formation Histories of Isolated Low-Mass Galaxies: Setting a New Empirical Benchmark

Special Requirements	Aperture PA Range 65 to 84 Degrees (V3 65.07457694 to 84.07457694) No Parallel Attachments Background Limited. Background no more than 10th percentile above minimum
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Proposal 9601 - Observation 4 - The Star Formation Histories of Isolated Low-Mass Galaxies: Setting a New Empirical Benchmark

Observation	Proposal 9601, Observation 4: AntliaB										Tue Apr 14 18:00:09 GMT 2026		
	Diagnostic Status: Warning												
	Observing Template: NIRCам Imaging												
	Coordinated Parallel Template(s): NIRISS Imaging												
Diagnostics	(AntliaB (Obs 4)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.												
	(Visit 4:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(4)	Antlia-B	RA: 09 48 57.0060 (147.2375250d)				Epoch of Position: 2000						
			Dec: -25 57 7.57 (-25.95210d)										
			Equinox: J2000										
			Comments: This object was generated by the targetselector and retrieved from the SIMBAD database. Category=Galaxy Description=[Dwarf galaxies, Field galaxies]										
Template	NIRCам Imaging					NIRISS Imaging							
	Module: ALL												
	Subarray: FULL												
	Target Placement: Module gap (large extended source)												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	NONE						1		4-POINT-SMALL-WITH-NIRISS		NO_DITHERING	
Spectral Elements	NIRCам Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F090W	F300M	MEDIUMDEEP8	7	3		12	4	12712.336			
	2	F150W	F335M	MEDIUMDEEP8	8	3		12	4	14644.954			
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	F090W		NIS	9	8		4	32	12712.336			
	2	F150W		NIS	9	8		4	32	12712.336			

Proposal 9601 - Observation 4 - The Star Formation Histories of Isolated Low-Mass Galaxies: Setting a New Empirical Benchmark

Special Requirements	Aperture PA Range 80.6 to 89 Degrees (V3 80.67457694 to 89.07457694) No Parallel Attachments Background Limited. Background no more than 10th percentile above minimum
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Proposal 9601 - Observation 3 - The Star Formation Histories of Isolated Low-Mass Galaxies: Setting a New Empirical Benchmark

Observation	Proposal 9601, Observation 3: UGC4879										Tue Apr 14 18:00:09 GMT 2026		
	Diagnostic Status: Warning												
	Observing Template: NIRCam Imaging												
	Coordinated Parallel Template(s): NIRISS Imaging												
Diagnostics	(UGC4879 (Obs 3)) Warning (Form): By selecting Target Placement = Module Gap the target coordinates will not fall on any detector unless an appropriate Mosaic, set of Dithers or Offset Special Requirement is specified.												
	(Visit 3:1) Warning (Form): Overheads are provisional until the Visit Planner has been run.												
Fixed Targets	#	Name	Target Coordinates				Targ. Coord. Corrections			Miscellaneous			
	(3)	UGC-4879	RA: 09 15 55.5387 (138.9814113d) Dec: +52 49 38.05 (52.82724d) Equinox: J2000				Epoch of Position: 2015.5						
	Comments: This object was generated by the targetselector and retrieved from the SIMBAD database.												
	Category=Galaxy												
	Description=[Dwarf galaxies, Field galaxies]												
Template	NIRCam Imaging					NIRISS Imaging							
	Module: ALL												
	Subarray: FULL												
	Target Placement: Module gap (large extended source)												
Dithers	#	Primary Dither Type		Primary Dithers		Dither Size		Subpixel Positions		Coordinated Parallel Subpixel Selector		Dither Direct Images Primes	
	1	NONE						1		4-POINT-SMALL-WITH-NIRISS		NO_DITHERING	
Spectral Elements	NIRCam Imaging	Short Filter	Long Filter	Readout Pattern	Groups/Int	Integrations/Exp		Total Integrations	Total Dithers	Total Exposure Time	Optional ETC ID		
	1	F090W	F300M	MEDIUMDEEP8	7	3		12	4	12712.336			
	2	F150W	F335M	MEDIUMDEEP8	8	3		12	4	14644.954			
Spectral Elements	NIRISS Imaging	Filter	Grism	Readout Pattern	Groups/Int	Integrations/Exp		Total Dithers	Total Integrations	Total Exposure Time	Optional ETC ID		
	1	F090W		NIS	9	8		4	32	12712.336			
	2	F150W		NIS	9	8		4	32	12712.336			

Proposal 9601 - Observation 3 - The Star Formation Histories of Isolated Low-Mass Galaxies: Setting a New Empirical Benchmark

Special Requirements	Aperture PA Range 133 to 145 Degrees (V3 133.07457694 to 145.07457694) No Parallel Attachments
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